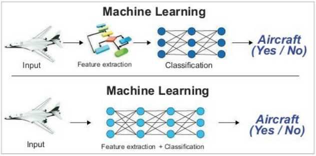

Figure 1: Differences between machine learning and deep learning

Table 1: Notable free and open source tools for deep learning

Tool/Library	URL
Apache Mahout	*https://mahout.apache.org/*
Apache Spark MLib	*https://spark.apache.org/mllib/*
Apache Singa	*https://singa.incubator.apache.org/en/index.html*
Chainer	*https://chainer.org/*
Caffe	*http://caffe.berkeleyvision.org/*
DLib	*http://dlib.net/*
DeepLearning4j	*https://deeplearning4j.org/*
Edward	*http://edwardlib.org/*
Gensim	*https://radimrehurek.com/gensim/*
Keras	*https://keras.io/*
Lime	*https://github.com/marcotcr/lime*
MXNet	*https://mxnet.apache.org/*
Microsoft Cognitive Toolkit	*https://www.microsoft.com/en-us/cognitive-toolkit/*
MLDB	*https://mldb.ai/*
Neon	*http://neon.nervanasys.com*
Oryx 2	*http://oryx.io/*
OpenAI	*https://openai.com/*
OpenNN	*http://www.opennn.net/*
PyTorch	*http://pytorch.org/*
Shogun	*http://www.shogun-toolbox.org/*
scikit-learn	*http://scikit-learn.org/*
TensorFlow	*https://www.tensorflow.org/*
Torch	*http://torch.ch/*

will detect only these two words and will respond as per the data it has been trained on. In the case of DL, the messages analogous to the training data will also be processed, like 'snatching', 'cannot see' and similar words. This is the beauty and power of DL – it enables machines to learn on their own, from the training data set.

As shown in Figure 1, the process of feature extraction from the input is a separate one in the case of machine learning. Researchers need to create a separate method for feature extraction in the case of ML, using different mathematical formulae and scientific computations. The methods and approaches for feature extraction are integrated in the case of deep learning, which makes it more accurate and smart.

DeepLearning4J: Deep learning in Java

DeepLearning4j (DL4J) *(https://deeplearning4j.org/)* is a Java based library that provides comprehensive and high performance implementations for DL in multiple applications. It is available as a free and open source deep learning library, with features like distributed computing for Java platforms, and can be used for knowledge discovery and deep predictive mining on graphics processing units (GPUs) as well as CPUs. The GPU is used for high performance and fast operations with a display of images, videos, animations and related multimedia objects.

Using Eclipse DeepLearning4j, Big Data analytics can be performed along with Apache Spark and Hadoop, using Java as well as Scala programming. DL4J integrates the techniques and algorithms of artificial intelligence (AI), which can be used for business intelligence, cyber forensics, robotic process automation (RBA), network intrusion detection and prevention, recommender systems, predictive analysis, regression, face recognition, natural language processing, anomaly detection and many others.

In addition to the inherent high performance features, DL4J enables the import of models from other prominent DL frameworks including TensorFlow, Theano, Caffe and Keras. DL4J provides the interface for both Java and Python programming without any compatibility issues.

The key features embedded in DeepLearning4j include the following:

- Scalability on Hadoop for Big Data
- Microservices architecture
- Parallel computing and training
- Massive amounts of data can be processed using clusters
- Support for GPUs for scalability on Amazon Web Services (AWS) cloud
- APIs for Java, Python and Scala
- Distributed architecture with multi-threading
- Support to CPUs and GPUs

DL4J has the following components and libraries for assorted applications, in order to support multiple functions.

- *ND4J:* This is the integration of NumPy for Java Virtual Machine (JVM). ND4J can be used to call the libraries for fast processing of matrix data on the processing units. It is a library for numerical computations and processing with performance-aware execution of multi-dimensional objects for scientific applications, including signal processing, linear algebra, optimisation, transformations,